Arthur Dexheimer Trein

Porto Alegre, Rio Grande do Sul, Brazil
+55 (51) 99239-5983

arthur.dexheimer@gmail.com arthurtrein.github.io

Research interests
Phonetics and phonology · L2 speech development · Speech perception and production · Quantitative methods

EDUCATION

2025–present
Porto Alegre, Brazil

MA in Language Studies (Psycholinguistics)
Universidade Federal do Rio Grande do Sul
Advisors: Ubiratã Alves† (UFRGS), Ingrid Finger (UFRGS) & Ronaldo Lima Jr. (UnB)
CAPES MA Fellowship (2026–2027)

> Thesis
> Investigating multilingual vowel development with Bayesian Generalized Additive Models

2020–2025
Porto Alegre, Brazil

BA in Languages and Literatures (Portuguese and English)
Universidade Federal do Rio Grande do Sul
Graduated with Academic Honors [Láurea Acadêmica]

> Thesis
> Métodos de análise de dados longitudinais de línguas não nativas via Teoria dos Sistemas Dinâmicos Complexos: um estudo do desenvolvimento, por aprendizes brasileiros, do padrão de *pre-fortis clipping* da língua inglesa
>
> Methods for analyzing longitudinal non-native language data through Complex Dynamic Systems Theory: A study of Brazilian learners' development of the English pre-fortis clipping pattern

PUBLICATIONS

Peer-reviewed journal articles

Under review — Jesus, T., Finger, I., Back, A. & **Trein, A.** Acurácia ortográfica em português de crianças bilíngues e monolíngues em alfabetização: um estudo longitudinal [Spelling accuracy in Portuguese among bilingual and monolingual children in early literacy: a longitudinal study]. Submitted to *Revista Brasileira de Linguística Aplicada*.

In press — **Trein, A.**, Alves, U., Kupske, F. & Gauer, L. Brazilian learners' development of English-L2 pre-fortis clipping patterns: a process-oriented study through Complex Dynamic Systems Theory. Submitted to *Revista da ABRALIN*.

2025 — Alves, U., **Trein, A.** & Gauer, L. Variability in the development of phonetic-phonological patterns by L2 learners: analyzing length gains in vowels preceding final voiced plosives in L2 English. *Letras de Hoje*, *60*(1), 1–15. doi:10.15448/1984-7726.2025.1.48439

Book chapters

In press — Alves, U., **Trein, A.** & Luchini, P. "Esse é o inglês que eu quero falar(?)(!)": atitudes de valoração social, por parte de aprendizes argentinos, frente a variedades "modelo" de inglês-L2 ["This is the English I want to speak(?)(!)": Argentine learners' social evaluation attitudes towards "model" varieties of L2 English]. In L. Brisolara & M. Brum-de-Paula (Eds.), *Fronteiras linguísticas: do chão ao coração*.

2026 — Alves, U., Milan, P., **Trein, A.**, Moraes, C., Vieira, F. & Alves, L. Development of speech fluency in an additional language: Analysis of breakdown rates in the speech of an Argentine learner of Brazilian Portuguese. In *Cuestiones del Lenguaje en Contextos Académicos y Profesionales: Enseñanza y Traducción*. Editorial de la Universidad Nacional de Mar del Plata.

Alves, U., Lima Jr., R. & **Trein, A.** Brazilian Students' Attitudes Towards Model Varieties of English: Implications for Learner Identity and L2 Pronunciation Development. In V. Sardegna, M. Lazzaro-Salazar & P. Luchini (Eds.), *(Mis)Understandings in Multicultural Communication: Implications for Second Language Classrooms and Professional Settings* (pp. 111–131). Springer.

2024 — **Trein, A.**, Alves, U. & Luchini, P. Accentedness perception ratings in non-native speaker-listener binomials: Analyzing the influence of proficiency. In P. Luchini, U. Alves & V. Innocentini (Eds.), *Otras cuestiones del lenguaje: traducción y enseñanza de lenguas extranjeras* (pp. 369–387). Universidad Nacional de Mar del Plata.

2022 **Trein, A.**, Alves, U. & Luchini, P. Comprehensibility ratings in non-native speaker-listener binomials: Implications for NNL teaching. In A. Sousa, R. Garcia & T. Santos (Eds.), *Perspectivas para o Ensino de Línguas* (Vol. 6, pp. 21–32). EDUFAC.

Other publications

2025 **Trein, A.** & Silva, J. Teaching English through the Olympic Games: a lesson plan for EJA students. *Revista Bem Legal*, *14*, 308–318. Teaching material

2024 **Trein, A.**, Vieira, F., Albuquerque, J., Pereira, L. & Alves, U. Módulo 1 – Nomes e apelidos [Module 1 – Names and nicknames]. *Sons do Braz[iw]* (ultrasound-based pronunciation teaching unit for Portuguese as an Additional Language). Teaching material

Trein, A. & Alves, U. O inglês britânico é mais inteligente? [Is British English more intelligent?]. *Jornal da Universidade* (UFRGS). Outreach

Trein, A. Orwell, língua e política: observações à luz da Análise do Discurso [Orwell, language and politics: observations in light of Discourse Analysis]. *Revista Letrar*, *3*, 143–152. Journal article

2023 Barros, A. & **Trein, A.** Second language learning motivation: an overview. *Revista Letrar*, *2*, 50–67. Journal article

Trein, A. Resenha de *Retrato do artista quando coisa*, de Manoel de Barros [Review of *Retrato do artista quando coisa*, by Manoel de Barros]. *Brasil/Brazil*, *36*(72). Book review

2022 **Trein, A.**, Vieira, F., Bitencourt, N. & Fernandes, J. Situação de crianças em zonas de guerra [The situation of children in war zones]. *UFRGSMUNDI*, *10*, 97–123. Study guide

2021 **Trein, A.**, Muller, B., Casanova, F., Constantino, G. & Ness, T. Proteção de florestas tropicais [Protection of tropical forests]. *UFRGSMUNDI*, *9*, 55–78. Study guide

More: 13 abstracts in conference proceedings; full list on Lattes.

PRESENTATIONS

Note: entries tagged PT were delivered in Portuguese; their titles are translated into English.

Talks

2026 **Trein, A.** Investigating linguistic development with Generalized Additive Models. *Ciclo de Palestras do I Encontro do Laboratório de Fonologia Não Nativa*. PT

Peer-reviewed conference presentations

2026 **Trein, A.** Effects of perceptual training and explicit pronunciation instruction on Brazilian learners' development of the pre-fortis clipping pattern in L2 English. *14º Interab - Congresso Internacional da ABRALIN*. Recife PT

Magagnin, K. & **Trein, A.** Pronunciation training via a mobile app: a process analysis of the effects of online digital resources on Brazilian English students' productions. *14º Interab - Congresso Internacional da ABRALIN*. Recife PT

Magagnin, K. & **Trein, A.** Pronunciation training via a mobile app: a process analysis with Bayesian Generalized Additive Mixed Models (GAMMs). *I Jornada Brasil-Portugal de Aquisição da Linguagem*. Pelotas PT

Trein, A., Vieira, F. & Alves, U. Modeling cross-linguistic speech influences over time through Generalized Additive Mixed Models: VOT production in Spanish, English and Brazilian Portuguese. *5th International Conference on Language Attrition and Bilingualism*. Porto Alegre Poster

Trein, A. & Alves, U. Influences of ultrasound use on an Argentine learner's vowel development in Brazilian Portuguese. *XVI Encontro do Círculo de Estudos Linguísticos do Sul (CELSUL)*. Porto Alegre PT

Magagnin, K. & **Trein, A.** Pronunciation training via a smartphone app: a process analysis of the effects of online digital resources on Brazilian English students' productions. *XVI Encontro do Círculo de Estudos Linguísticos do Sul (CELSUL)*. Porto Alegre PT

Trein, A. & Alves, U. Pronunciation teaching with ultrasound: a study of an Argentine learner's vowel development in Brazilian Portuguese. *I Jornada Brasil-Portugal de Aquisição da Linguagem*. Pelotas PT

2026 Jesus, T., **Trein, A.**, Finger, I. & Back, A. Early writing development in Spanish-Portuguese bilingual children: cross-linguistic influences and repertoire dynamics. *5th International Conference on Language Attrition and Bilingualism*. Porto Alegre Poster

2025 **Trein, A.**, Alves, U., Lima Jr., R. & Gauer, L. Influences of perceptual training and explicit pronunciation instruction on L2 phonetic-phonological development: a study of Brazilian learners' acquisition of English pre-fortis clipping. *III Fonapliando: Encontro de fonética, fonologia e suas interfaces*. Florianópolis PT

2024 **Trein, A.** Complex-dynamic measures of variability and their role in product analyses: Brazilian learners' duration gains in vowels preceding [b d g] in L2 English. *V Seminário Internacional de Fonologia*. Rio de Janeiro Poster PT

Trein, A., Vieira, F. & Pereira, L. Keeping an eye on speech: an ultrasound-based teaching unit for pronunciation in Portuguese as an Additional Language. *V Seminário Internacional de Fonologia*. Rio de Janeiro Poster PT

2023 **Trein, A.** & Alves, U. Accentedness perception in speaker-listener pairs of English as a lingua franca. *I Seminário de Linguística de Contato*. Porto Alegre PT

Milan, P., **Trein, A.**, Moraes, C., Vieira, F. & Alves, L. Investigating oral fluency development: breakdown rates in an Argentine learner's productions in Portuguese as an Additional Language. *XIII Congresso Internacional da Abralin*. Curitiba PT

Presentations in regional and institutional events

2025 **Trein, A.** Analyzing non-native pronunciation development with Bayesian GAMMs: a study of Brazilian learners' acquisition of pre-fortis clipping in L2 English. *I Jornada do Observatório da Fala: bilinguismo, fonologia e sociedade*. Porto Alegre PT

Trein, A. The role of explicit instruction and perceptual training in L2 phonetic-phonological development: a study of Brazilian learners' acquisition of the pre-fortis clipping pattern in L2 English. *II Psycholink*. Porto Alegre PT

2024 **Trein, A.** & Alves, U. Variability in language learners' acquisition of phonetic-phonological patterns: analyzing duration gains in vowels preceding [b d g] in L2 English. *I PsychoLink*. Porto Alegre PT

Trein, A. Complex-dynamic measures of variability and their role in product analyses: Brazilian learners' duration gains in vowels closed by [b d g] in L2 English. *V Jornada de Educação Linguística e Literária*. Porto Alegre PT

Trein, A., Moraes, C., Vieira, F., Alves, L., Alves, U. & Milan, P. Speech fluency development in an additional language: breakdown rates in an Argentine learner's Portuguese productions. *V Encontro do Grupo de Pesquisa NUPFFALE*. Florianópolis PT

Trein, A., Alves, U. & Lima Jr., R. Investigating social evaluation: Brazilian English language professionals' evaluations of American and British English. *V Encontro do Grupo de Pesquisa NUPFFALE*. Florianópolis PT

Trein, A. & Alves, U. Investigating Brazilian English language professionals' attitudinal evaluations of model varieties of the non-native language. *II Jornada Leitura, Educação e suas ressonâncias*. Online PT

2023 **Trein, A.** Brazilian English language professionals' attitudinal evaluations of native varieties of English. *XXIV Salão de Iniciação Científica da PUCRS*. Porto Alegre PT

Trein, A. & Alves, U. Brazilian English language professionals' attitudinal judgments of native varieties of English. *XXXV Salão de Iniciação Científica da UFRGS* (regular and special sessions). Porto Alegre PT

2022 **Trein, A.** Accentedness perception in a lingua franca English context: on the role of speakers and listeners. *XXIII Salão de Iniciação Científica da PUCRS*. Porto Alegre PT

Trein, A. Accentedness perception in non-native speaker-listener binomials. *XXXIV Salão de Iniciação Científica da UFRGS*. Porto Alegre PT

Trein, A. & Alves, U. Pronunciation models of Brazilian English language professionals: implications for teaching. *Fonologia e Ensino: Descobertas e Interfaces*. Online PT

Trein, A. & Alves, U. Brazilian language and literature students' attitudinal evaluations of native varieties of English. *II NUPFFALE Online Meeting*. Online PT

Trein, A. Linguistic identities of English learners from Rio Grande do Sul: an analysis based on the L2 Motivational Self System. *I Seminário de Pesquisas em Fonética e Fonologia do Laboratório de Bilinguismo e Cognição*. Porto Alegre PT

2021 **Trein, A.** Comprehensibility ratings in a non-native binomial: an emergent construct. *XXXIII Salão de Iniciação Científica da UFRGS*. Online PT

RESEARCH EXPERIENCE

2025–present **MA Student Researcher**

Bilingualism and Cognition Lab & Non-Native Phonology Lab

Supervisors: Ubiratã Alves & Ingrid Finger

Modeling strategies for L2 speech development data

Model longitudinal change and cross-linguistic influences in multilingual speech (e.g. VOT and vowel production) with Bayesian Generalized Additive Mixed Models (GAMMs)

- Measure and annotate speech production data in *Praat*
- Visualize data and fit Bayesian GAMMs in *R* (*brms*)

Output: MA thesis (in progress) · 1 manuscript in preparation · 1 talk · 3 oral presentations · 1 poster

Optimizing Perception Learning

Big-team, international collaborative project on High Variability Phonetic Training

- Collect speech data for the Brazilian site
- Annotate and analyse speech data in *Praat*

Early L2 literacy development

Develop a standardized test battery of morphological skills and analyze early literacy data

- Take part in project meetings to design and refine the battery
- Visualize data and fit frequentist Generalized Linear Mixed Models (GLMMs) in *R*

Output: 1 manuscript under review · 1 poster

2021–2025 **Undergraduate Research Assistant**

Bilingualism and Cognition Lab & Non-Native Phonology Lab

Supervisor: Ubiratã Alves

L2 speech development under High Variability Phonetic Training and explicit instruction

Tracked how learners' production of target L2 acoustic features developed during pronunciation training

- Measured and annotated speech production data in *Praat*
- Visualized and analyzed the experimental data

Output: BA thesis · 2 journal articles · 6 oral presentations · 1 poster

Ratings of L2 English speech

Examined how Brazilian and Argentine learners and teachers of English rate L2 speech for comprehensibility, accentedness and social evaluation

- Designed and conducted listener rating experiments
- Analyzed rating data and discussed implications for pronunciation teaching

Output: 4 book chapters · 10 oral presentations · 1 outreach article

Other collaborative projects

- *Speech fluency in L2 Brazilian Portuguese*: annotated breakdown and repair measures in *Praat* and analyzed how fluency changed over time
- *Ultrasound-assisted pronunciation instruction*: co-developed an ultrasound-based teaching unit, delivered the instruction and collected ultrasound and acoustic data

Output: 1 book chapter · 4 oral presentations · 1 poster · 1 teaching unit

FUNDING AND AWARDS

Fellowships and scholarships

2026–2027 **MA Fellowship**. *CAPES* (Brazilian Federal Agency for Support and Evaluation of Graduate Education).

2023–2025 **Undergraduate Research Scholarship**. *CNPq* (Brazilian National Council for Scientific and Technological Development).

2022–2023 **Undergraduate Research Scholarship**. *Universidade Federal do Rio Grande do Sul*.

2021–2022 **Undergraduate Research Scholarship**. *FAPERGS* (Research Support Foundation of the State of Rio Grande do Sul).

Awards and honors

2025 **Academic Honors** [Láurea Acadêmica], BA in Languages and Literatures. *Universidade Federal do Rio Grande do Sul*.

2024 **Honorable Mentions**, poster sessions (two posters). *V Seminário Internacional de Fonologia*.

2023 **Best Paper Award** in Linguistics and Language Studies. *XXIV Salão de Iniciação Científica da PUCRS*.

Honorable Mention, UFRGS Young Researcher Award 2023 in Language Studies, Linguistics and Arts. *XXXV Salão de Iniciação Científica da UFRGS*.

2022 **Best Paper Award** in Linguistics and Language Studies. *XXIII Salão de Iniciação Científica da PUCRS*.

Session Distinction (session: Language and Cognition). *XXXIV Salão de Iniciação Científica da UFRGS*.

TEACHING EXPERIENCE

Ongoing **Graduate Teaching Assistant**
Complementary Studies of English I (English phonetics and phonology), UFRGS
Instructors: Ubiratã Alves & Ingrid Finger

- Prepare and teach lectures for undergraduate students

2021 **Undergraduate Teaching Assistant**
English III (English grammar), UFRGS
Instructor: Ubiratã Alves

- Corrected assignments, provided feedback and supported students online with course content

ACADEMIC SERVICE

Conference organization

2026 *I Jornada Brasil-Portugal de Aquisição da Linguagem*. Organizing committee. Pelotas

2024 *I PsychoLink*. Organizing committee. Porto Alegre

2023 *I Seminário de Linguística de Contato*. Organizing committee. Porto Alegre

Seminário Pesquisa em Movimento. Organizing committee. Porto Alegre

2022 *Fonologia e Ensino: Descobertas e Interfaces*. Organizing committee. Online

Academic journals

2023–present **Editorial board member**. *Organon* (UFRGS).

2023–2025 **Peer reviewer**. *Revista Letrar* (ISSN 2764-9962), issues 3–6.

Other service

2020–2022 **Academic Team Member**, *Model United Nations conferences* at UFRGS: study guides, chairing and delegate training.

SKILLS

Languages	Portuguese (*native*), English (*near-native*), Spanish (*basic*)
Statistics	Frequentist and Bayesian Generalized Linear Mixed Models (GLMMs) and Generalized Additive Mixed Models (GAMMs)
Programming	*R* (data wrangling, data visualization and statistical analysis)
Software	*Praat* (speech analysis), *Audacity* (audio editing), *Typst*, LaTeX and *Quarto* (authoring)